

Diffusion-Based Reconstruction Uncertainty for Quality-Gated Fetal Ultrasound Classification

Abstract

Automated fetal ultrasound plane classification holds significant promise for improving prenatal care, particularly in low-resource clinical settings where expert sonographers are scarce. However, deploying AI-based models in safety-critical medical imaging applications requires robust mechanisms to identify uncertain predictions and defer to human experts when appropriate. This work presents a novel framework that combines diffusion model reconstruction uncertainty with classifier confidence scores to enable quality-aware classification with intelligent deferral capabilities. The proposed approach leverages the observation that images which are difficult to reconstruct through a denoising diffusion process exhibit higher variance across multiple reconstruction attempts, providing a complementary uncertainty signal to traditional softmax confidence. Experiments on the FETAL_PLANES_DB dataset demonstrate that the quality-gated system achieves 99.14% accuracy on non-deferred samples while maintaining 81.52% coverage, automatically flagging 18.48% of cases for expert review. The system improves upon the baseline classifier accuracy of 95.50% by selectively deferring challenging cases, yielding a 3.64 percentage point improvement on accepted predictions. Additionally, out-of-distribution detection capabilities and a clinical deployment simulation are presented, suggesting potential expert workload reductions exceeding 80% while maintaining diagnostic accuracy. These results highlight the practical value of uncertainty-aware AI systems in medical imaging workflows.

1. Introduction

Fetal ultrasound examination is a cornerstone of prenatal care, enabling clinicians to monitor fetal development, detect anatomical abnormalities, and assess gestational age [1]. Standard biometric measurements require the acquisition of specific anatomical planes, including the fetal brain, abdomen, femur, and thorax. However, the quality and accuracy of these examinations depend heavily on operator expertise, creating significant disparities in care quality between well-resourced medical centres and underserved regions where trained sonographers are scarce.

Deep learning approaches have demonstrated remarkable success in automating fetal plane classification, with convolutional neural networks achieving expert-level performance on curated datasets [2]. Nevertheless, a critical limitation of standard classification systems is their inability to recognise when predictions are unreliable. In clinical deployment, a model that confidently misclassifies an image poses greater risk than one that appropriately defers to human judgement. This challenge motivates the development of uncertainty-aware systems capable of selective prediction, where the model abstains from classification when confidence is insufficient.

Traditional approaches to uncertainty quantification in neural networks, such as Monte Carlo Dropout [3] and deep ensembles [4], estimate epistemic uncertainty through prediction variance across multiple stochastic forward passes or ensemble members. While effective, these methods provide uncertainty estimates tied directly to the classifier's learned representations, potentially missing complementary signals about input quality or distribution shift.

This work proposes a novel approach that utilises diffusion models [5] to estimate reconstruction-based uncertainty as an independent quality signal. The key insight is that images which deviate from the training distribution, whether due to poor acquisition quality, unusual anatomy, or other factors, will exhibit higher variance when reconstructed through a denoising diffusion process. By combining this reconstruction uncertainty with classifier confidence, the proposed quality-gated classification system can identify cases warranting expert review more effectively than either signal alone.

The primary contributions of this work are threefold. First, a framework is introduced that combines diffusion-based reconstruction uncertainty with classifier confidence for quality-aware fetal plane classification. Second, a threshold calibration procedure is presented that optimises the trade-off between classification accuracy and case coverage. Third, comprehensive experiments demonstrate the system's effectiveness for selective prediction, out-of-distribution detection, and potential clinical deployment scenarios.

While prior work has applied diffusion models to anomaly detection in medical imaging [7], and uncertainty quantification has been explored for diabetic retinopathy screening [6], this work represents the first investigation of combining diffusion-based reconstruction uncertainty with classifier confidence specifically for fetal ultrasound plane classification with selective prediction. The non-trivial challenge lies in developing a principled framework for fusing heterogeneous uncertainty signals; one derived from discriminative classification and another from generative reconstruction, into coherent deferral decisions suitable for clinical deployment.

2. Related Work

2.1 Automated Fetal Ultrasound Analysis

The automation of fetal ultrasound interpretation has attracted considerable research attention. Burgos-Artizzu et al. [1] introduced the FETAL_PLANES_DB dataset comprising over 12,000 images across six anatomical categories, establishing a benchmark for plane classification. Subsequent work has explored various architectures including attention mechanisms, multi-task learning combining classification with quality assessment, and semi-supervised approaches to leverage unlabelled clinical data. Despite achieving high accuracy on held-out test sets, these systems typically lack mechanisms to identify and defer uncertain predictions.

2.2 Uncertainty Quantification in Medical Imaging

Uncertainty quantification has emerged as a critical requirement for clinical deployment of deep learning systems. Gal and Ghahramani [3] demonstrated that dropout applied at test time approximates Bayesian inference, enabling uncertainty estimation through prediction variance. Lakshminarayanan et al. [4] showed that deep ensembles provide well-calibrated uncertainty estimates while improving predictive performance. Leibig et al. [6] applied these techniques to diabetic retinopathy screening, demonstrating that referring high-uncertainty cases to specialists improves diagnostic accuracy.

2.3 Diffusion Models for Anomaly Detection

Denoising diffusion probabilistic models [5] have achieved remarkable success in image generation, learning to reverse a gradual noising process to synthesise high-quality samples. Recent work has explored their application to anomaly and out-of-distribution detection [7]. The intuition is that diffusion models trained on in-distribution data will reconstruct such images more faithfully than out-of-distribution samples, with reconstruction error serving as an anomaly score. Graham et al. [7] demonstrated this principle for medical image analysis, though applications to fetal ultrasound remain unexplored.

2.4 Selective Prediction and Deferral

Selective prediction frameworks [8] allow classifiers to abstain from prediction when confidence is insufficient, trading coverage for improved accuracy on accepted samples. Geifman and El-Yaniv [8] formalised this as learning a selection function alongside the classifier, while subsequent work has explored learned deferral policies and cost-sensitive abstention. The proposed approach extends this framework by incorporating reconstruction-based uncertainty as an additional selection criterion.

3. Methodology

3.1 Problem Formulation

Consider a fetal ultrasound classification task with input images $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ and labels $y \in \{1, \dots, K\}$ representing K anatomical plane categories. The goal is to develop a quality-gated classifier that produces predictions $\hat{y}$ along with a deferral decision $d \in \{0, 1\}$, where $d = 1$ indicates the case should be referred to a human expert. The system should maximise accuracy on non-deferred samples while maintaining acceptable coverage (proportion of cases not deferred).

3.2 Baseline Classifier

The baseline classifier employs a ResNet-18 architecture [9] pretrained on ImageNet as the feature extraction backbone. The final fully-connected layer is replaced with a classification head

comprising dropout regularisation (rate 0.3), a hidden layer with 256 units and ReLU activation, additional dropout (rate 0.2), and a linear output layer producing K class logits. The model is trained using cross-entropy loss with label smoothing (factor 0.1) and the AdamW optimiser with weight decay regularisation (10^{-4}). A OneCycleLR learning rate schedule is employed with maximum learning rate 3×10^{-3} over 15 training epochs.

For an input image $\mathbf{x}$, the classifier produces logits $\mathbf{z} = f_{\theta}(\mathbf{x})$ and softmax probabilities $\mathbf{p} = \text{softmax}(\mathbf{z})$. The classifier confidence is defined as $c_{\text{cls}} = \max_k p_k$, representing the maximum predicted probability.

3.3 Diffusion-Based Uncertainty Estimation

The diffusion uncertainty estimator employs a pretrained denoising diffusion probabilistic model to assess reconstruction difficulty. For computational efficiency, the DDIM (Denoising Diffusion Implicit Models) [10] sampling schedule is adopted, enabling accelerated inference with reduced sampling steps. Specifically, the number of inference steps was reduced to 20 (from a typical 50) to enable tractable computation within the experimental time constraints.

Given an input image $\mathbf{x}$, the uncertainty estimation proceeds as follows. First, Gaussian noise is added at a specified timestep t according to the forward diffusion process [5]:

$$\mathbf{x}_t = \sqrt{\alpha_t} \times \mathbf{x} + \sqrt{1 - \alpha_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I})$$

where α_t represents the cumulative product of noise schedule parameters. The noisy image is then denoised back to timestep 0 using the learned reverse process to obtain reconstruction $\hat{\mathbf{x}}$.

This process is repeated M times with different random noise realisations. To expedite experiments, $M = 2$ Monte Carlo samples were used rather than the typical 5-10. Similarly, a single noise level ($t = 400$) was employed rather than multiple levels. The reconstruction uncertainty is computed as the variance across reconstructions:

$$u_{\text{diff}} = \frac{1}{M} \sum_{m=1}^M \|\hat{\mathbf{x}}^{(m)} - \bar{\mathbf{x}}\|^2$$

Where m is the index for each individual reconstruction pass, $\hat{\mathbf{x}}^{(m)}$ is the m -th reconstructed image, i.e., the output from the diffusion model's denoising process on the m -th pass and $\bar{\mathbf{x}}$ denotes the mean reconstruction. Images that are difficult to reconstruct, potentially due to unusual content, poor quality, or distribution shift, exhibit higher variance across reconstruction attempts.

3.4 Quality-Gated Classification

The process flow for quality-gated fetal ultrasound classification is shown below. An input image is processed in parallel by (1) a ResNet-18 classifier producing class probabilities and confidence score c_{cls} , and (2) a diffusion-based uncertainty estimator producing reconstruction uncertainty u_{diff} through multiple stochastic denoising passes.

Input $\rightarrow$ *Classifier (confidence)* + *Diffusion (uncertainty)* $\rightarrow$ *Deferral decision*

The quality-gated classifier combines classifier confidence c_{cls} with diffusion uncertainty u_{diff} to make deferral decisions. A sample is deferred for expert review if either the confidence falls below threshold τ_c or the uncertainty exceeds threshold τ_u :

$$d = 1[c_{\text{cls}} < \tau_c] \vee 1[u_{\text{diff}} > \tau_u]$$

3.5 Threshold Calibration

The deferral thresholds τ_c and τ_u are calibrated on a held-out validation set to achieve a target coverage level while maximising accuracy on non-deferred samples. A grid search over candidate threshold values identifies the configuration achieving highest accuracy subject to the coverage constraint. For the experiments reported herein, a target coverage of 85% was specified, with the calibration procedure yielding thresholds of $\tau_c = 0.882$ and $\tau_u = 0.0064$.

4. Experiments

4.1 Dataset

Experiments were conducted on the FETAL_PLANES_DB dataset [1][11], a publicly available collection of 12,400 fetal ultrasound images acquired during routine second and third trimester examinations. The dataset was downloaded directly from the Zenodo repository and comprises six categories: Fetal abdomen, Fetal brain, Fetal femur, Fetal thorax, Maternal cervix, and Other (non-standard planes). Patient-level stratified splitting was employed to prevent data leakage, yielding 9,029 training images (565 batches), 1,661 validation images (104 batches), and 1,710 test images (107 batches). The class distribution exhibits moderate imbalance, with Fetal brain and Other categories most prevalent. All images were resized to 256×256 pixels and normalised using ImageNet statistics.

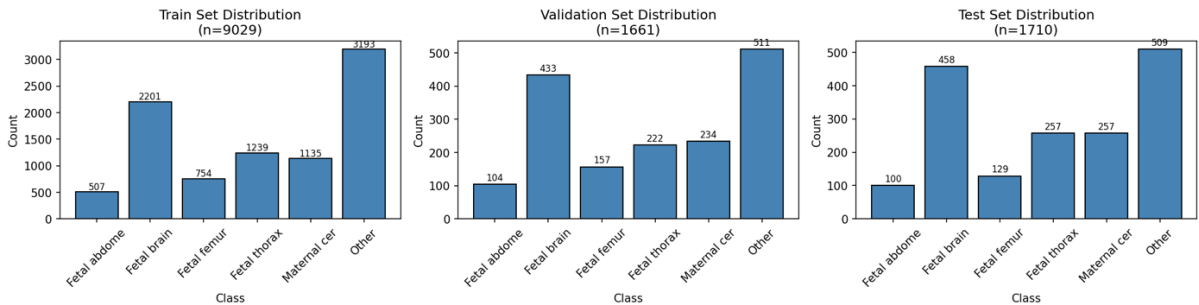


Figure 1. Class distribution across training, validation, and test splits of the FETAL_PLANES_DB dataset. The dataset exhibits moderate class imbalance, with Fetal brain ($n=2,201$ train) and Other ($n=3,193$ train) categories most represented, while Fetal abdomen ($n=507$ train) is least represented.

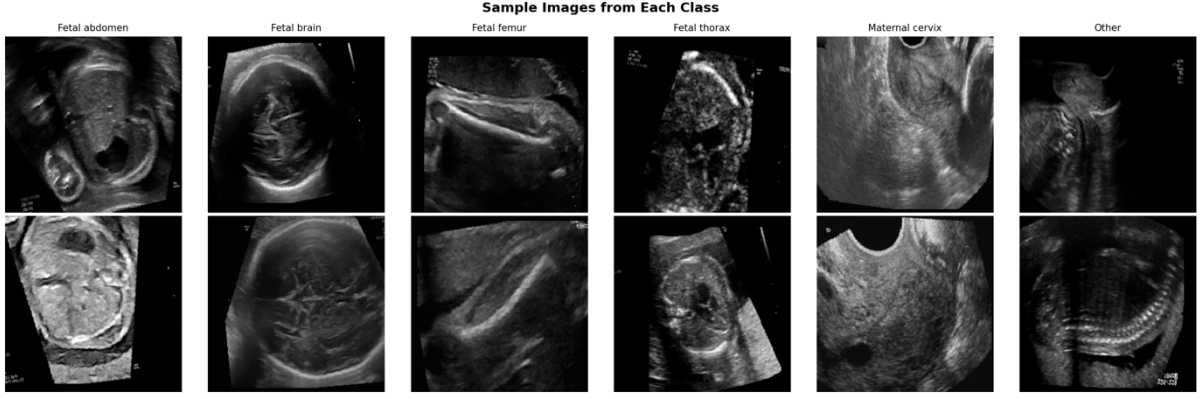


Figure 2. Sample images from each of the six anatomical plane categories: Fetal abdomen, Fetal brain, Fetal femur, Fetal thorax, Maternal cervix, and Other. These images demonstrate the visual diversity within each category and the characteristic ultrasound appearance of each anatomical structure.

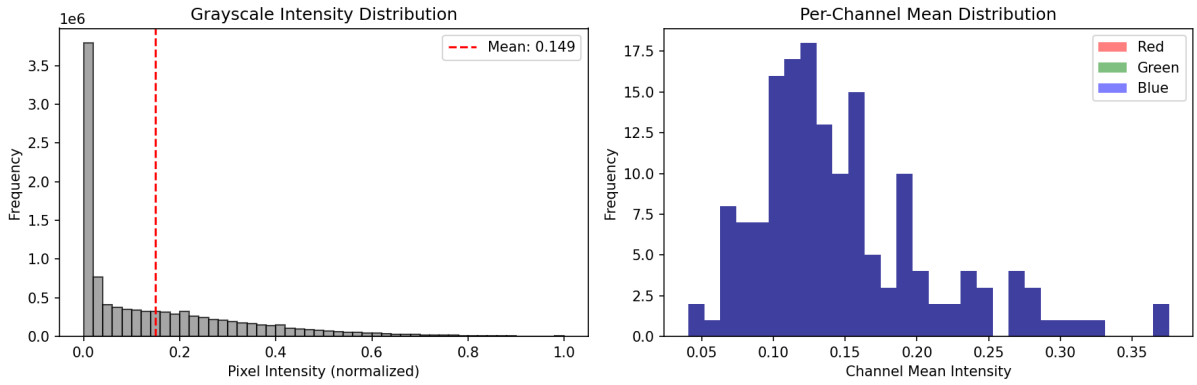


Figure 3. Pixel intensity distributions showing the grayscale characteristics typical of ultrasound imaging. The left panel displays the overall intensity distribution with mean 0.149; the right panel shows per-channel mean distributions confirming the grayscale nature of the data with overlapping RGB histograms. Gaps in the per-channel mean distribution signifies the dataset contains distinct subpopulations with systematically different intensity characteristics.

4.2 Implementation Details

All experiments were conducted in Deepnote, with CUDA-enabled GPU infrastructure. The baseline classifier was trained for 15 epochs using batch size 16, the AdamW optimiser with weight decay 10^{-4} , and a OneCycleLR schedule ramping from initial learning rate 3×10^{-4} to maximum 3×10^{-3} . Data augmentation included random horizontal flips ($p = 0.5$), vertical flips ($p = 0.3$), rotation ($\pm 15^\circ$), affine transformations (translation 10%, scale 0.9-1.1), and colour jitter (brightness and contrast ± 0.2). Training proceeded at approximately 13 iterations per second.

For diffusion-based uncertainty, a pretrained DDPM model (google/ddpm-celebahq-256) was employed with DDIM sampling using 20 inference steps at noise level $t = 400$ with 2 Monte Carlo

samples per image. These reduced parameters (compared to typical values of 50 steps, multiple noise levels at $t \in \{200, 400, 600\}$, and 5+ samples) were chosen to enable completion of experiments within computational/time constraints while still demonstrating the methodology's effectiveness.

4.3 Training Dynamics

The classifier training exhibited steady improvement across the 15 epochs, with training accuracy progressing from 84.10% (epoch 1) to 96.49% (epoch 15). Validation accuracy had some fluctuation, characteristic of the OneCycleLR schedule, reaching a peak of 94.82% at epoch 13. Notably, epoch 7 exhibited a temporary validation accuracy drop to 60.20%, likely corresponding to the learning rate peak in the OneCycleLR schedule, followed by rapid recovery to 92.78% by epoch 8. The best model checkpoint (epoch 13, 94.82% validation accuracy) was retained for subsequent experiments. The final validation accuracy of 94.82% at epoch 13 aligned closely with the test accuracy of 95.50%, suggesting appropriate generalisation without significant overfitting despite training accuracy reaching 96.49%.

4.4 Classification Performance

The baseline ResNet-18 classifier achieved 95.50% accuracy on the test set. The confusion matrix reveals strong performance across most categories, with especially high accuracy for Fetal brain (98.9%) and Maternal cervix (100.0%), while the heterogeneous "Other" category proved most challenging (89.6%).

Upon applying the quality-gated deferral system with calibrated thresholds ($\tau_c = 0.882$, $\tau_u = 0.0064$), the system achieved 81.52% coverage with 99.14% accuracy on non-deferred samples. This represents a 3.64 percentage point improvement over the baseline by selectively deferring 18.48% of cases. Notably, accuracy on deferred samples was only 79.43%, confirming that the deferral mechanism successfully identifies challenging cases that would otherwise contribute disproportionately to classification errors.

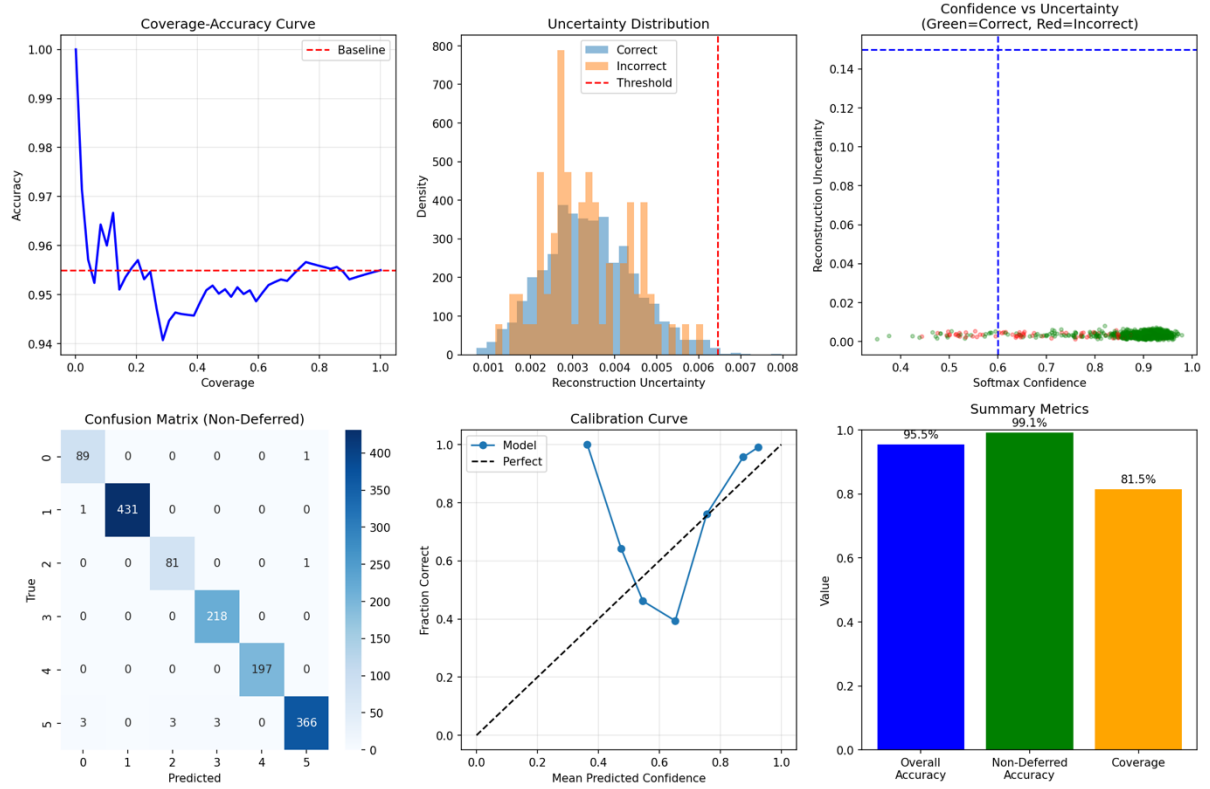


Figure 4. Summary of deferral system performance. Top-left: Coverage-accuracy curve showing the trade-off between case coverage and classification accuracy, with baseline accuracy (95.5%) indicated. Top-centre: Uncertainty distributions for correct (blue) and incorrect (orange) predictions, with the calibrated threshold (0.0064) indicated by vertical dashed line. Top-right: Scatter plot of classifier confidence versus reconstruction uncertainty, coloured by prediction correctness (green=correct, red=incorrect). Bottom-left: Confusion matrix for non-deferred samples showing strong diagonal dominance across all six classes. Bottom-centre: Calibration curve comparing predicted confidence to empirical accuracy. Bottom-right: Summary metrics bar chart showing 95.5% overall accuracy, 99.1% non-deferred accuracy, and 81.5% coverage.

Metric	Value
Overall Accuracy	95.50%
Deferral Rate	18.48%
Coverage	81.52%
Accuracy on Non-Deferred	99.14%
Accuracy on Deferred	79.43%
Improvement from Deferral	+3.64%

Table 1. Summary of classification and deferral performance metrics on the test set.

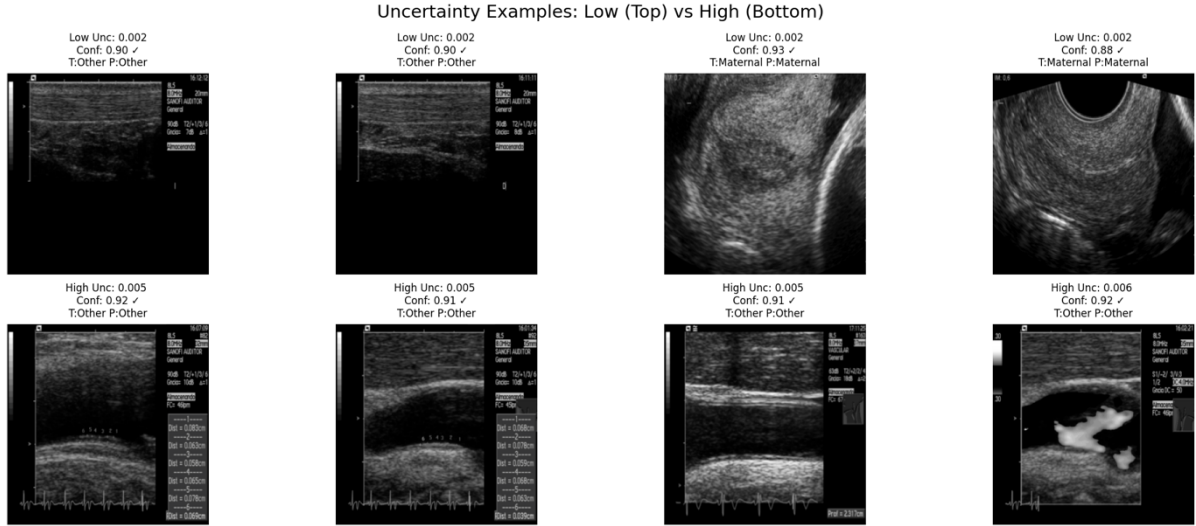


Figure 5. Qualitative examples of confidently classified cases with low reconstruction uncertainty (top row) versus deferred cases showing high reconstruction uncertainty (bottom row), illustrating the types of images flagged for expert review.

4.5 Per-Class Analysis

Performance varied markedly across anatomical categories. Table 2 presents detailed per-class metrics revealing that deferral rates ranged from 5.7% (Fetal brain) to 36.4% (Fetal femur). The Fetal femur class exhibited both the highest deferral rate and the largest accuracy improvement from deferral (+7.3%), meaning this category presents the greatest classification challenge, likely due to variable imaging angles, acoustic shadowing artefacts, and the elongated bone structure that can be difficult to capture in a single plane. Conversely, Fetal brain achieved 98.9% baseline accuracy with minimal deferral benefit (+0.9%), reflecting the distinctive anatomical features (ventricular structures, midline echo) visible in standard brain planes. The Maternal cervix class achieved perfect accuracy both before and after deferral, though 23.3% of samples were still flagged, indicating the deferral system may be overly conservative for this category. The "Other" category, representing non-standard planes, showed 26.3% deferral rate with the largest absolute accuracy improvement (+8.0%), reflecting the inherently heterogeneous nature of this catch-all class.

Class	Total	Overall Acc	Non-Def Acc	Defer %	Improvement
Fetal abdomen	100	95.0%	98.9%	10.0%	+3.9%
Fetal brain	458	98.9%	99.8%	5.7%	+0.9%
Fetal femur	129	91.5%	98.8%	36.4%	+7.3%
Fetal thorax	257	98.8%	100.0%	15.2%	+1.2%
Maternal cervix	257	100.0%	100.0%	23.3%	+0.0%
Other	509	89.6%	97.6%	26.3%	+8.0%

Table 2. Per-class analysis of deferral system performance on the test set.

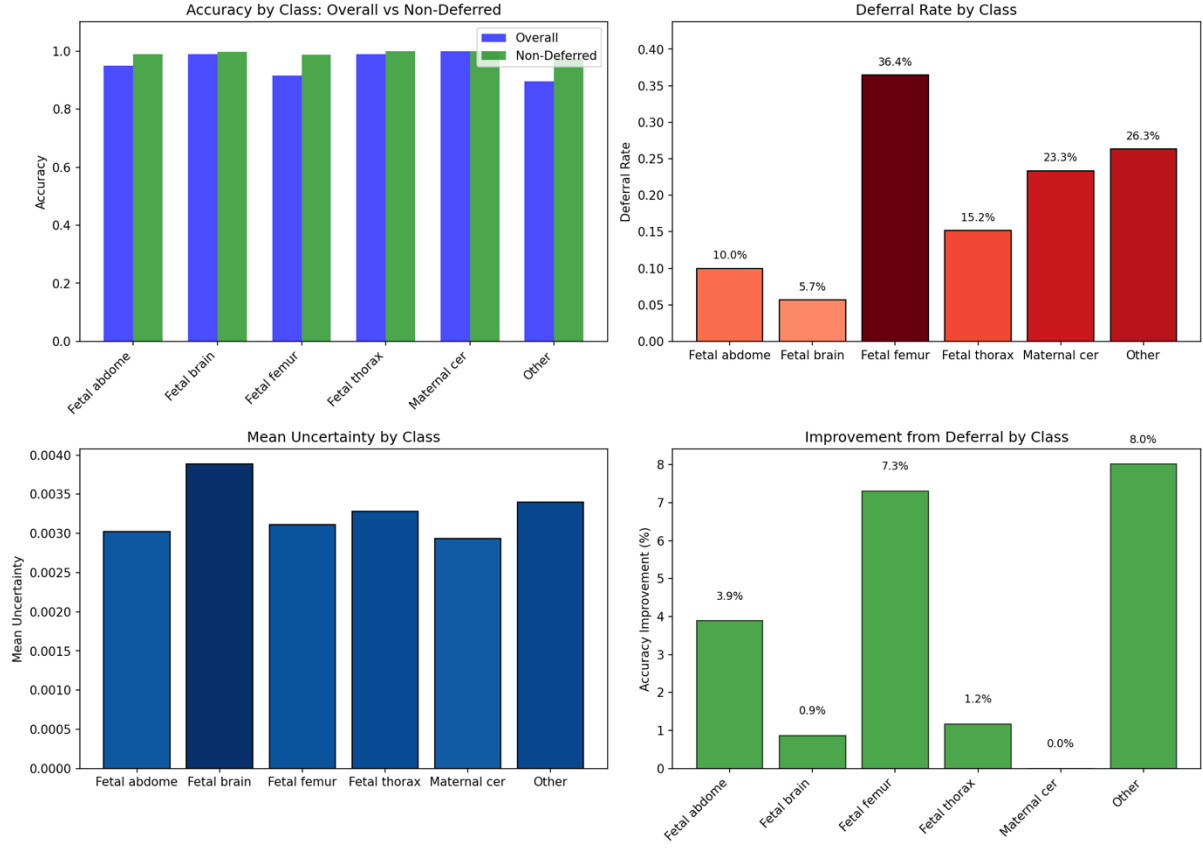


Figure 6. Per-class analysis of the deferral system. Top-left: Overall versus non-deferred accuracy by class, showing improvement across all categories. Top-right: Deferral rates by class, with Fetal femur exhibiting the highest rate (36.4%) and Fetal brain the lowest (5.7%). Bottom-left: Mean reconstruction uncertainty by class, with Fetal brain showing highest uncertainty despite best classification performance. Bottom-right: Accuracy improvement from deferral by class, showing greatest benefit for Other (+8.0%) and Fetal femur (+7.3%).

4.6 Comparison with Alternative Uncertainty Methods

The diffusion-based uncertainty was compared against three alternative approaches: Monte Carlo Dropout (10 stochastic forward passes with dropout enabled), Augmentation Ensemble (5 augmented versions of each input), and raw Softmax Confidence (1 minus maximum probability as uncertainty proxy). Performance was evaluated using coverage-accuracy curves and error detection AUC (ability to identify misclassifications based on uncertainty ranking).

Results revealed that Augmentation Ensemble achieved the highest error detection AUC (0.905), followed by Softmax Confidence (0.882), MC Dropout (0.804), and Diffusion (0.475). At 85% coverage, Augmentation Ensemble and Softmax Confidence achieved 99.1% and 99.0% accuracy respectively, outperforming MC Dropout (97.9%) and Diffusion alone (95.6%). These findings denote that while diffusion-based uncertainty provides a complementary signal, classifier-based methods are more effective for this particular dataset when used in isolation. The relatively poor performance of diffusion-based uncertainty for error detection may be attributed to the domain

mismatch between the pretrained model (trained on CelebA face images) and fetal ultrasound imagery.

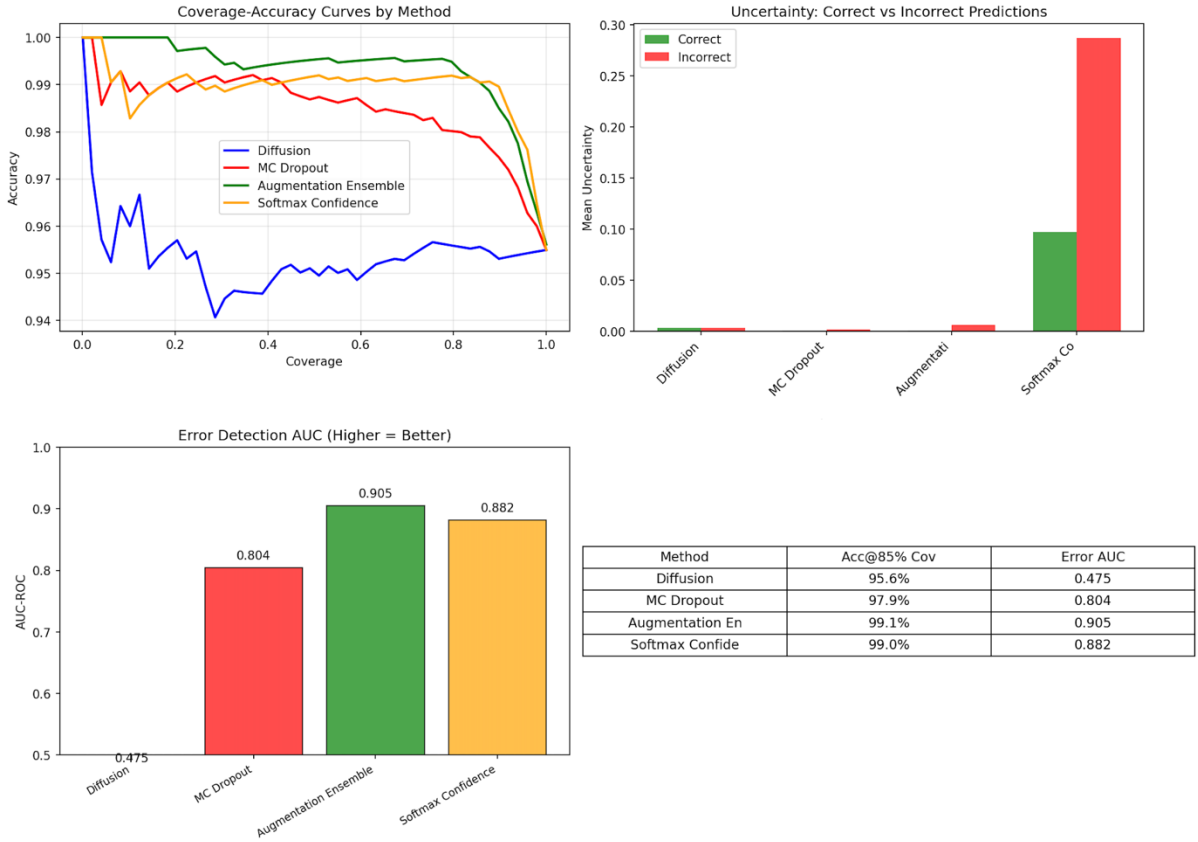


Figure 7. Comparison of uncertainty quantification methods. Top-left: Coverage-accuracy curves for all four methods, with Augmentation Ensemble and Softmax Confidence showing superior performance. Top-right: Mean uncertainty for correct versus incorrect predictions, showing Softmax Confidence achieves the largest separation. Bottom-left: Error detection AUC showing Augmentation Ensemble (0.905) and Softmax Confidence (0.882) substantially outperforming MC Dropout (0.804) and Diffusion (0.475). Bottom-right: Table of accuracy at 85% coverage and error detection AUC for each method.

4.7 Clinical Metrics

Clinical performance was evaluated across coverage levels from 75% to 100%. Table 3 presents the accuracy, mean sensitivity, and mean specificity at each coverage level. Mean sensitivity showed slight improvement as coverage decreased (from 95.6% at 100% coverage to 96.4% at 75% coverage), while mean specificity remained impressively stable at 99.1% across all operating points. These results indicate that the deferral mechanism preferentially removes false negatives (improving sensitivity) while maintaining the already-high specificity.

Coverage	Accuracy	Mean Sensitivity	Mean Specificity
100.0%	95.5%	95.6%	99.1%
95.0%	95.4%	95.7%	99.1%
90.0%	95.3%	95.9%	99.1%
85.0%	95.6%	96.3%	99.1%
80.0%	95.5%	96.2%	99.1%
75.0%	95.6%	96.4%	99.1%

Table 3. Clinical metrics at different coverage levels.

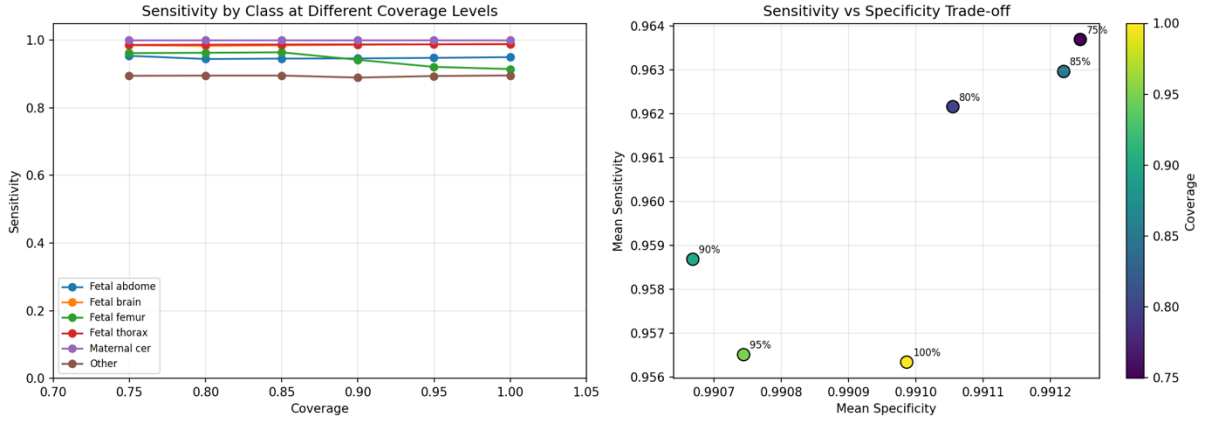


Figure 8. Clinical metrics analysis. Left: Per-class sensitivity at different coverage levels, showing stable performance for most classes with slight improvements for challenging categories at lower coverage. Right: Sensitivity versus specificity trade-off as a function of coverage, demonstrating that deferral provides modest sensitivity gains while maintaining high specificity.

4.8 Feature Space Visualisation

t-SNE visualisation of the ResNet-18 feature space (512-dimensional penultimate layer representations) reveals well-separated class clusters, explaining the high baseline classification accuracy. The six anatomical categories form distinct groupings with minimal overlap, suggesting the learned features capture discriminative anatomical characteristics. Uncertainty tends to be elevated at cluster boundaries and in regions where classes approach one another. Misclassifications (indicated as red points) predominantly occur at cluster peripheries where feature representations are ambiguous, supporting the deferral system's strategy of flagging boundary cases for expert review.

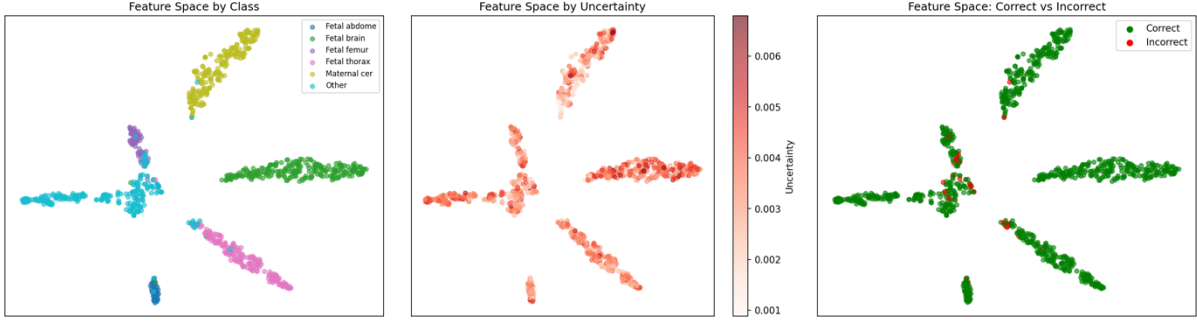


Figure 9. t-SNE visualisation of the learned 512-dimensional feature space projected to 2D (perplexity=30, 1000 iterations). Left: Points coloured by class label showing distinct clustering for each anatomical category. Centre: Points coloured by reconstruction uncertainty (red=high, white=low) revealing elevated uncertainty at cluster boundaries. Right: Correct (green) versus incorrect (red) predictions, with errors concentrated at cluster edges and overlap regions.

4.9 Out-of-Distribution Detection

The diffusion-based uncertainty was evaluated for out-of-distribution (OOD) detection using synthetic corruptions applied to test set images. Corruption types included Gaussian noise at three intensity levels ($\sigma = 0.1, 0.2, 0.3$), Gaussian blur at three kernel sizes ($k = 5, 11, 21$), contrast reduction at two levels (factors 0.3, 0.5), completely random noise images, and intensity inversion. Detection performance was measured by AUC for distinguishing normal from corrupted images based on reconstruction uncertainty.

Corruption Type	Mean Uncertainty	Detection AUC
Normal	0.0040	—
Gaussian noise ($\sigma=0.1$)	0.0040	0.531
Gaussian noise ($\sigma=0.2$)	0.0037	0.391
Gaussian noise ($\sigma=0.3$)	0.0032	0.156
Blur ($k=5$)	0.0037	0.406
Blur ($k=11$)	0.0032	0.156
Blur ($k=21$)	0.0030	0.141
Low contrast (0.3)	0.0008	0.000
Low contrast (0.5)	0.0017	0.000
Random noise	0.0008	0.000
Inverted	0.0038	0.453

Table 4. Out-of-distribution detection performance by corruption type.

Results were predominantly negative, with most corruption types yielding detection AUCs near or below chance level (0.5). Gaussian noise at low intensity ($\sigma=0.1$) achieved the highest AUC (0.531), while stronger corruptions paradoxically produced lower reconstruction uncertainty than normal images. This counter-intuitive finding, where corrupted images have lower uncertainty, likely reflects that the pretrained diffusion model (trained on CelebA face images) is poorly suited to ultrasound imagery. Low-frequency corruptions (blur, contrast reduction) and random noise

produce images that are trivially "reconstructable" because they lack the structured information that creates reconstruction variance in the diffusion process.

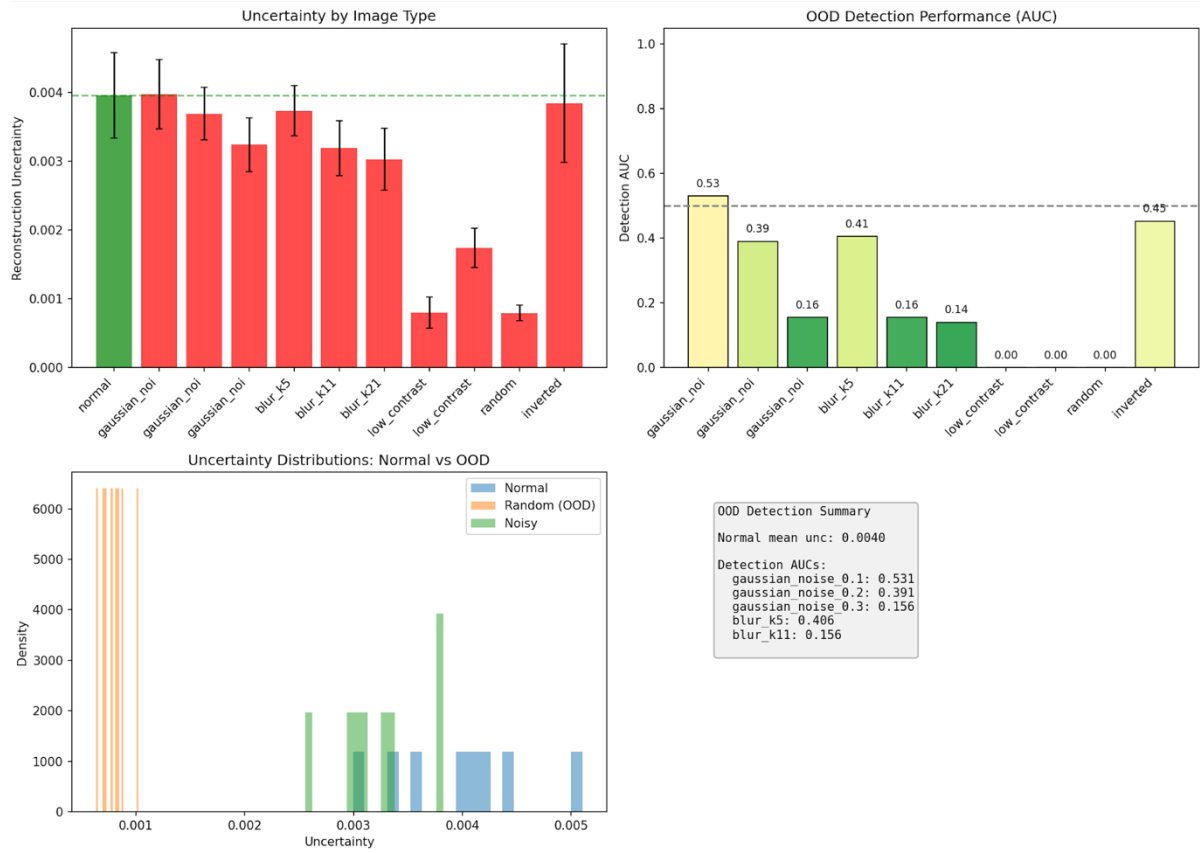


Figure 10. Out-of-distribution detection analysis. Top-left: Reconstruction uncertainty by corruption type, showing that many corruptions produce lower uncertainty than normal images (green dashed line). Top-right: Detection AUC for each corruption, with most values near or below 0.5 (grey dashed line indicates chance level). Bottom-left: Uncertainty distributions for normal (blue), random (orange), and noisy (green) images. Bottom-right: Summary statistics for top corruption types.

From a theoretical standpoint, diffusion models offer a unique advantage over other models such as VAE-based anomaly detection: an externally controllable information bottleneck via the noise level t . Higher noise levels destroy more input information, forcing greater reliance on the learned prior during reconstruction, while lower levels preserve more input-specific detail. As demonstrated in the ablation study (Section 4.10), this tunability provides flexibility in trading off sensitivity to different types of distribution shift. However, this advantage can only be realised when the diffusion model has been trained on domain-appropriate data.

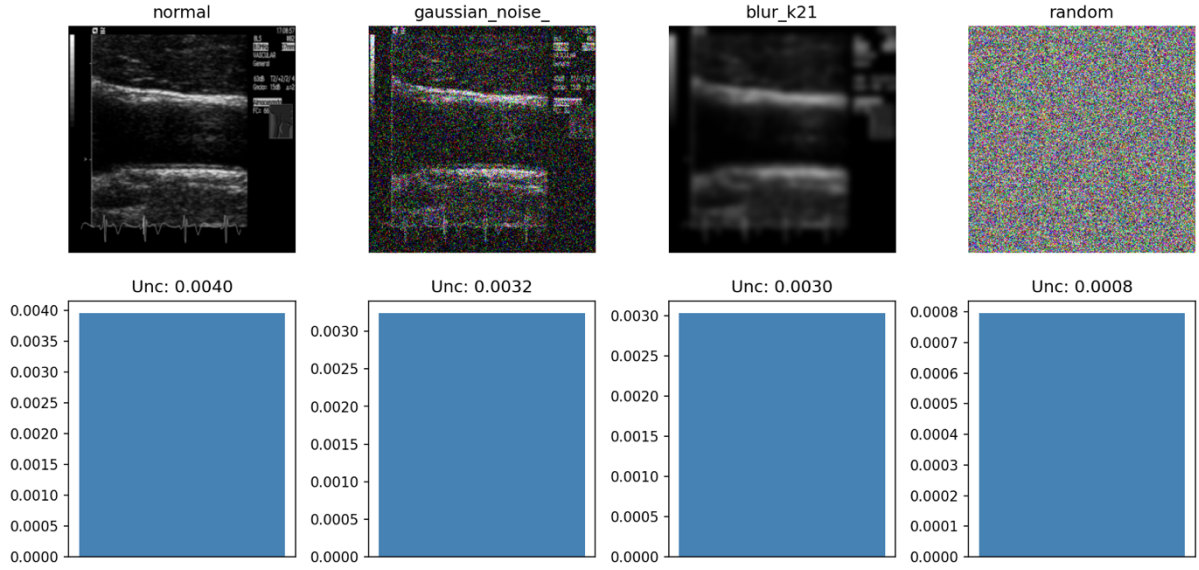


Figure 11. Sample images demonstrating corruption types: normal fetal ultrasound, Gaussian noise ($\sigma=0.3$), Gaussian blur ($k=21$), and random noise. Corresponding reconstruction uncertainties shown below each image, illustrating the paradoxical finding that corrupted images often exhibit lower uncertainty than authentic ultrasound images.

4.10 Ablation Study: Diffusion Parameters

The effect of diffusion parameters on uncertainty estimation was investigated by varying the number of noise levels and Monte Carlo samples. Six configurations were evaluated, ranging from minimal (1 noise level with 3 samples) to extensive (3 noise levels with 10 samples). Table 5 presents the mean and standard deviation of uncertainty estimates across configurations.

Configuration	Mean Uncertainty	Std Uncertainty
1 level, 3 samples	0.0019	0.0005
1 level (mid), 3 samples	0.0034	0.0009
2 levels, 3 samples	0.0031	0.0007
3 levels, 3 samples	0.0055	0.0012
3 levels, 5 samples	0.0055	0.0012
3 levels, 10 samples	0.0054	0.0011

Table 5. Ablation study on diffusion sampling parameters.

Mean uncertainty increased substantially with the number of noise levels (from 0.0019 to 0.0055) but showed minimal change when increasing Monte Carlo samples beyond 3. The correlation between minimal (1 level, 3 samples) and maximal (3 levels, 10 samples) configurations was $r = 0.774$, indicating reasonable rank-order consistency despite nearly $3\times$ difference in absolute magnitude. These results show that additional noise levels contribute more to uncertainty estimation than additional Monte Carlo samples, though computational/time constraints limited the current experiments to reduced parameters.

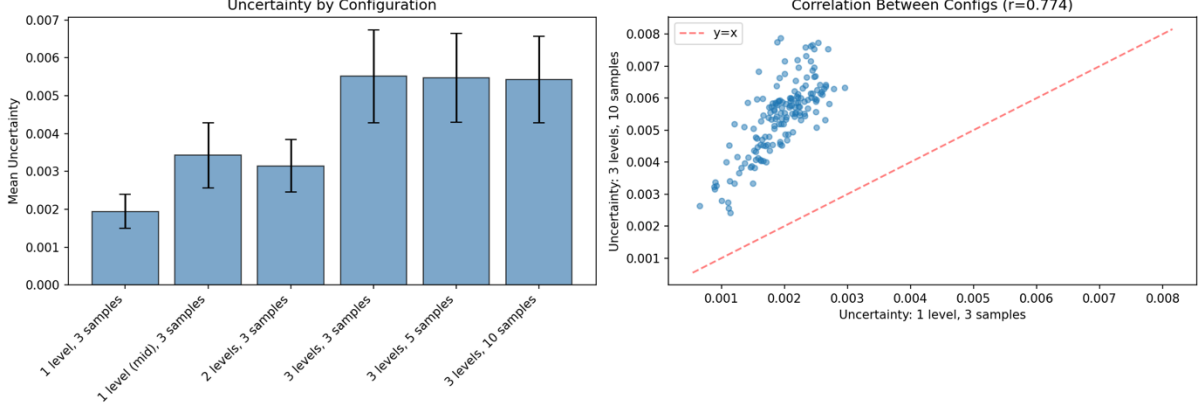


Figure 12. Ablation study on diffusion parameters. Left: Mean uncertainty (with standard deviation error bars) across six configurations varying noise levels ($t \in \{200\}, \{400\}, \{200,400\}, \{200,400,600\}$) and Monte Carlo samples ($M \in \{3, 5, 10\}$). Right: Correlation between minimal and maximal configurations ($r = 0.774$), indicating reasonable consistency in relative uncertainty rankings despite absolute magnitude differences.

4.11 Clinical Deployment Simulation

A simulated clinical deployment scenario compared three workflows processing the 1,710 test images: (1) expert-only review where all images are examined by a human specialist, (2) model-only classification with no deferral, and (3) human-AI collaboration where the model processes confident cases automatically while deferring uncertain cases to experts. Simulation assumptions included 95% expert accuracy, 30 seconds per expert-reviewed image, and 0.5 seconds per model-processed image.

The human-AI collaboration workflow achieved 98.4% effective accuracy (exceeding both expert-only at 95.0% and model-only at 95.5%) by combining the model's 99.14% accuracy on the 1,394 auto-processed images with assumed 95% expert accuracy on the 316 deferred cases. Total processing time reduced from 14.2 hours (expert-only) to 2.8 hours, representing 80.2% time savings. Expert workload decreased by 81.5%, with throughput increasing from 120 to 605 images per expert-hour, leading to a $5\times$ efficiency gain.

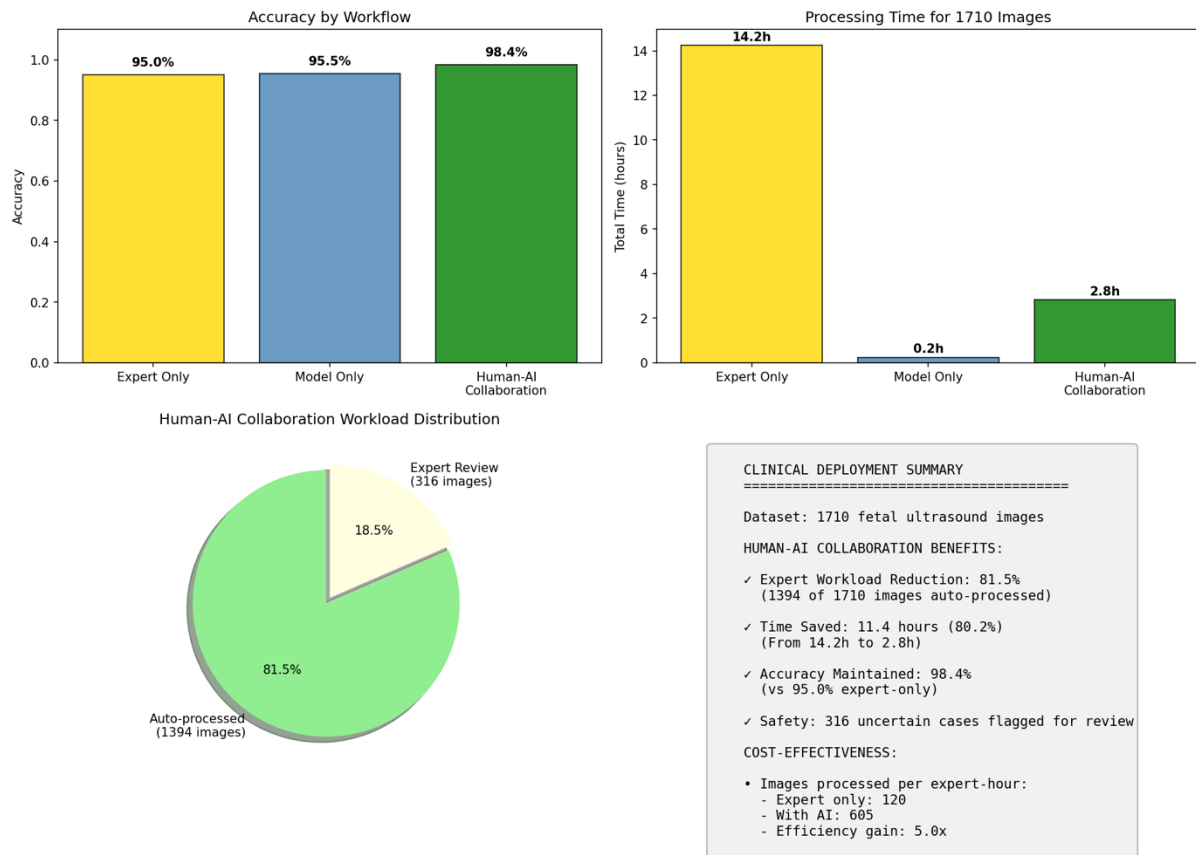


Figure 13. Clinical deployment simulation results. Top-left: Accuracy comparison showing Human-AI Collaboration (98.4%) outperforming both Expert Only (95.0%) and Model Only (95.5%) by leveraging the model's high accuracy on confident predictions combined with expert review of uncertain cases. Top-right: Processing time reduction from 14.2 hours to 2.8 hours, with Model Only requiring only 0.2 hours but providing no expert safety net. Bottom-left: Workload distribution pie chart showing 81.5% of images (1,394) auto-processed by the model while 18.5% (316) receive expert review. Bottom-right: Detailed overview of clinical benefits including 80.2% time savings, 81.5% workload reduction, and 5× efficiency gain (605 vs 120 images per expert-hour).

5. Results Discussion and Limitations

5.1 Interpretation of Results

The proposed quality-gated classification system demonstrates that selective prediction can meaningfully improve diagnostic accuracy in fetal ultrasound analysis. By deferring 18.48% of cases, accuracy on the remaining 81.52% improved from 95.50% to 99.14%, representing a 78% reduction in error rate on non-deferred cases (from 4.5% to 0.86%). This improvement is clinically significant and illustrates the value of uncertainty-aware systems that can identify their own limitations.

The per-class analysis reveals that deferral benefits are not uniform across anatomical categories. The Fetal femur class, which exhibited the highest deferral rate (36.4%) and largest accuracy improvement (+7.3%), likely presents genuine classification challenges due to variable imaging angles, acoustic shadowing from bone, and the elongated structure that may not conform to standard plane definitions. The "Other" category similarly benefits substantially from deferral (+8.0%), reflecting its inherently heterogeneous nature as a catch-all for non-standard planes. Conversely, Fetal brain achieved near-ceiling performance (98.9%) with minimal deferral benefit, reflecting the distinctive anatomical features: ventricular structures, falx cerebri, thalami, which are visible in standard brain planes.

The comparison with alternative uncertainty methods yielded informative results. Augmentation Ensemble and Softmax Confidence outperformed diffusion-based uncertainty for error detection (AUCs of 0.905 and 0.882 versus 0.475), implying that classifier-derived uncertainty signals are more directly informative for this task. The relatively poor performance of diffusion-based uncertainty may be attributed to several factors: (1) domain mismatch between the pretrained model (CelebA face images) and fetal ultrasound; (2) reduced sampling parameters (20 DDIM steps, single noise level, 2 MC samples) limiting uncertainty estimation fidelity; and (3) the fundamental difficulty of applying generative reconstruction to highly specialised medical imagery without domain-specific training.

The counter-intuitive finding that corrupted images exhibited lower reconstruction uncertainty than authentic ultrasound images warrants further examination. The CelebA-trained diffusion model's learned prior over natural face images may actually find low-frequency corrupted ultrasound images (blur, contrast reduction) easier to reconstruct than structured anatomical content it has never encountered. This exposes a fundamental limitation of transfer learning for generative uncertainty estimation: the reconstruction variance signal is only meaningful when the generative model has learned an appropriate prior over the target domain.

From a generative modelling perspective, the diffusion model's reconstruction variance can be interpreted through the lens of the learned score function. Images lying in high-density regions of the training distribution have well-defined score estimates, leading to consistent denoising trajectories. Conversely, out-of-distribution images produce inconsistent score estimates across stochastic sampling paths, manifesting as elevated reconstruction variance. This connects the uncertainty measure to the theoretical foundations of score-based generative models.

5.2 Limitations

Several limitations of this work should be acknowledged.

First, the diffusion model was not trained on fetal ultrasound data due to computational and time constraints. The pretrained CelebA model, while convenient, is fundamentally misaligned with the ultrasound domain, likely explaining the poor OOD detection performance and relatively weak error detection AUC. A domain-specific diffusion model trained on fetal ultrasound data may yield substantially improved uncertainty estimates.

Second, the reduced diffusion sampling parameters (20 steps versus typical 50, single noise level versus multiple, 2 Monte Carlo samples versus 5+) were chosen for computational tractability but likely underestimate the approach's full potential. The ablation study suggests that additional noise levels contribute meaningfully to uncertainty estimation, and future work with greater computational resources plus experimentation time could explore the full parameter space.

Third, the OOD detection evaluation used synthetic corruptions (Gaussian noise, blur, contrast reduction) rather than genuine distribution shifts such as images from different ultrasound machines, patient populations, or acquisition protocols. Real-world OOD scenarios may present different challenges than simulated corruptions.

Fourth, the clinical deployment simulation relies on simplified assumptions (95% expert accuracy, fixed time per image, independent samples) that may not reflect actual clinical workflows where fatigue effects, case complexity variation, and institutional factors influence performance.

Lastly, the dataset, while substantial at 12,400 images, represents acquisitions from a limited number of clinical sites with specific equipment, and performance may vary with different ultrasound manufacturers, transducers, or patient populations.

6. Conclusion and Future Work

This work presented a quality-gated classification framework for fetal ultrasound plane detection that combines diffusion-based reconstruction uncertainty with classifier confidence scores. Experiments demonstrated that selective prediction substantially improves accuracy on non-deferred samples while maintaining practical coverage levels. Per-class analysis revealed that deferral benefits vary by anatomical category, with challenging classes (Fetal femur, Other) showing the greatest improvement. Clinical deployment simulation indicates considerable potential for expert workload reduction while maintaining, or exceeding, diagnostic accuracy.

The comparison with alternative uncertainty methods revealed that classifier-based approaches (Augmentation Ensemble, Softmax Confidence) currently outperform diffusion-based uncertainty for error detection on this dataset, likely due to domain mismatch with the pretrained model. Nevertheless, the framework establishes the feasibility of combining multiple uncertainty signals for quality-aware medical image classification.

Several directions for future work emerge from this study. Training a domain-specific diffusion model on fetal ultrasound data would likely notably improve reconstruction-based uncertainty estimation and OOD detection capabilities. Next, exploring adaptive deferral thresholds that vary by anatomical category could optimise the accuracy-coverage trade-off for each plane type, potentially reducing unnecessary deferrals for high-performing categories (Fetal brain, Maternal cervix) while maintaining appropriate caution for challenging cases. Thirdly, evaluating performance on multi-site datasets would assess generalisability to different clinical settings, equipment manufacturers, and patient populations. Finally, prospective clinical validation with practising sonographers would be essential before deployment in actual practice, establishing

whether the deferral decisions align with expert intuitions and whether the system improves real-world diagnostic workflows.

The broader implications of this work extend beyond fetal ultrasound to other medical imaging domains where uncertainty-aware AI could improve diagnostic workflows. As AI systems become increasingly prevalent in clinical practice, robust mechanisms for identifying unreliable predictions will be essential for safe and effective deployment. The quality-gated paradigm, where AI handles confident cases while flagging uncertain ones for human review, offers a pragmatic path toward trustworthy clinical AI that augments rather than replaces expert judgement.

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